Bike-sharing demand: what 50,000 trips can and cannot tell us

SQL (MySQL) and Power BI analysis · Parth Dhawan · October 2026

7 tables · 8 SQL scripts · 3-page Power BI dashboard · Coding Ninjas case study

Flat demand, and no commute story in this data

138 vs 136

trips a day, weekend vs weekday

Weekends hold 29% of trips only because there are fewer of them. Per day they are busier.

3,461

trips imply over 40 km/h on a bike

With 72 over-capacity stations and 899 repairs dated before purchase, the data looks generated.

32

stations gain or lose 40+ bikes

Station net flow, revenue mix and repair types still give usable answers.

Seven tables, eight SQL scripts, one dashboard



1. Load

7 CSVs into MySQL with one build script

2. Check

Nulls, duplicates, durations + 7 reality checks

3. Explore

Dimensions, date ranges and measures

4. Analyse

Demand, segments, maintenance, revenue

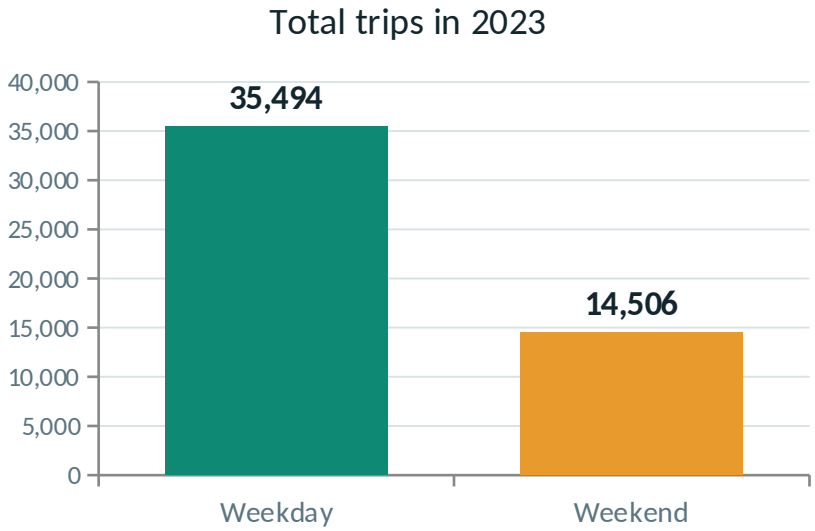
5. Report

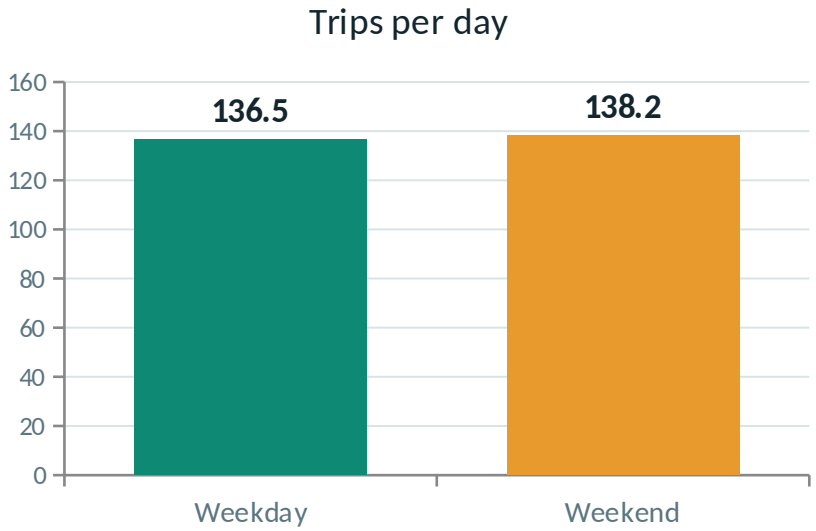
3-page Power BI dashboard

The data: 50,000 trips · 5,000 riders · 200 stations · 1,000 bikes · 9,998 repairs · 5 US cities · all of 2023.

Business question from the brief: where and when to position bikes, how to balance stations, and where maintenance is needed.

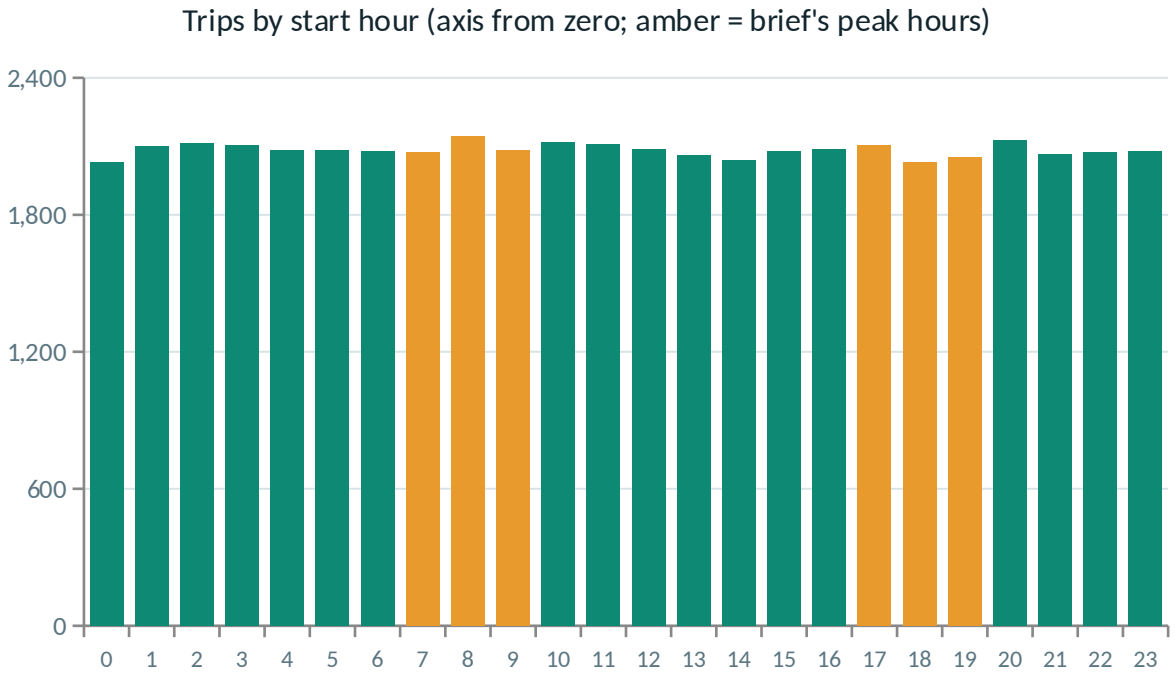
Weekends look quieter, but per day they are busier





2023 has 260 weekdays and 105 weekend days. The 2.4× gap in totals is the calendar, not commuters.

No rush hour: trips barely change across the day



5.5%

gap between the busiest hour (8 AM) and the quietest

158 / 200

stations peak outside commute hours (75% by chance)

32.2 vs 31.9

minutes per trip, peak vs off-peak

Four checks show the data is generated, not observed

72 of 200

stations report more bikes available than they can hold

3,461

trips imply over 40 km/h, up to 120 km/h; distance and duration correlate at 0.001

899

repairs are dated before the bike was bought; 996 of 1,000 last-service dates disagree with the log

1 city / hour

weather is logged for one city at a time, so it cannot explain city demand

Clean on the surface: 0 duplicate trips, 0 missing values, 0 negative durations. 2 duplicate repair rows removed.

Three answers the data can still support

Rebalancing

+65 / −63

Station 1 gains 65 more bikes than it loses; Station 148 loses 63. 32 stations are off by 40 or more.

Revenue mix

60.3%

of 136,065 in fares comes from subscriptions. February's −15.2% dip is −6.1% per day: mostly its 28 days.

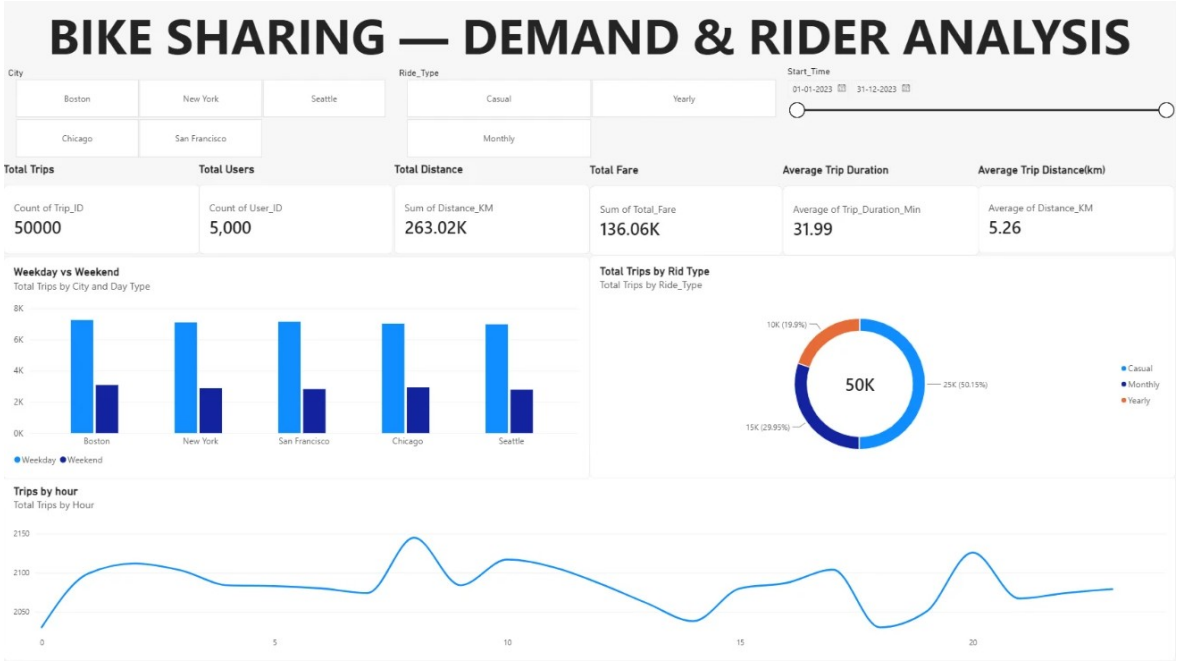
Maintenance

r = 0.04

between a bike's trips and its repairs. Flat tyres lead with 2,586 of 9,998 repairs.

None of these depend on realistic trip speeds or timing, so they hold even on generated data.

A 3-page Power BI dashboard on the MySQL tables



1. Overview

KPIs, weekday vs weekend by city, ride types, trips by hour

2. Operations

Station peaks, issues by bike type and season, age and gender

3. Revenue growth

Monthly revenue and month-on-month growth

Slicers for city, ride type and date. Arrivals use an inactive relationship with USERELATIONSHIP.

Next steps: rebalance now, collect real data before planning

1 Rebalance the 32 most imbalanced stations on a daily route, starting with the biggest gainers and losers.

2 Report demand per day and per station-hour, never as raw totals across uneven periods.

3 Collect dock-sensor counts, GPS distance and odometer readings, then rerun the commute and maintenance analysis.

4 Fix the source: cap available bikes at capacity and block repair dates before purchase dates.

Happy to walk you through the SQL

Every number in this deck is reproduced by the repo's build and checked on every push.

github.com/parthdhawan921-ops/bike-sharing-demand-analysis

parthdhawan921-ops.github.io/Parth_Personal/#bike

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